



Focus on innovation.

We take care of the rest.

Smart machines and accessories for prototyping, run by an AI engineer.

proto plans every step. baker makes the board. **mixer** will test it.

protopus-kappa.vercel.app

A prototype is a loop, and every lap costs weeks



Nobody is short of ideas. **They are short of tries.**

AI already builds software. In hardware, it stops at the file.

SOFTWARE

\$100M ARR

Lovable, 8 months after launch,
selling to people who can't code.

HARDWARE FILES

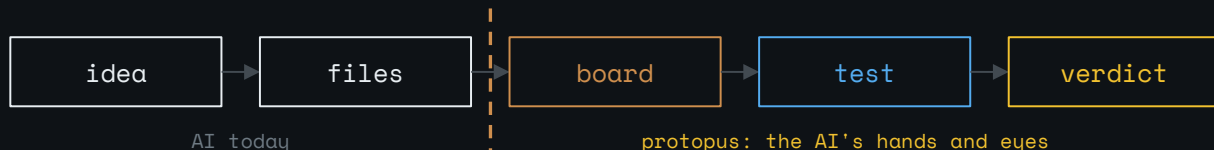
~\$100M

raised by AI that draws boards: Flux
\$49M, Quilter \$40M, Diode \$11.4M.
Claude drives Fusion and Blender (Apr
2026).

THE BENCH

none yet

an AI that can make the board, power
it and tell you if it works.



Smart machines give the AI
access to the real world.

Every rung of hardware knowledge loses people. Proto removes rungs.

RUNG	YOU USED TO NEED	WHO CARRIES IT NOW
01 idea → spec	What exactly should it do?	Proto asks, then writes it
02 parts	Datasheets, stock, footprints	Proto picks in-stock parts
03 schematic	Circuit theory, references	Proto + Flux check it
04 layout	An EDA tool, DRC	Proto fixes it (Baker Studio)
05 fabrication	Vendors, customs, 7-20 days	Baker, in hours
06 assembly	SMD soldering, stencil, reflow	Baker + printed jig
07 bring-up	Scope, logic analyser, firmware	Mixer measures (step 2)
08 debug	Root cause, next revision	Proto explains the fix

rungs 01-04: software can carry them · rungs 05-08: they need machines

HOW PEOPLE LEARN

By doing: try, look, adjust (Kolb's cycle). The speed of feedback sets the speed of learning.

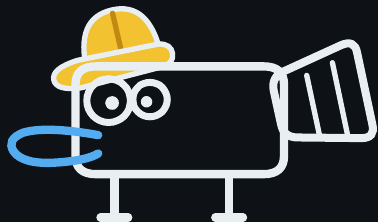
7-20 days

per try today: beginners quit, experts batch changes

≈2 hours

per try on Baker (target): many more attempts in one semester

One AI engineer, and the smart machines it runs



STEP 1 · NOW

proto

Plans the build, drives the machines, explains the results.



STEP 1 · NOW

baker

Makes the board: mill, mask, stencil, paste, reflow.



STEP 2 · AFTER POC

mixer

Tests it: eight instruments on one cable.

STEP 3

Probe Matrix: probes under the board, so Proto can reach, test and program every net. Then any 3D printer for the enclosure.

FOR PROTOTYPES · the first 1-10 boards of a design. Volume goes to a fab. DEMOS · [website](#) · Baker prototype video

A 3D printer doesn't replace the mould. Baker doesn't replace the fab.

THE PATH FROM IDEA TO 1,000s

EXISTS

a plastic bottle

PROTOTYPE desktop 3D print:
shape, feel, fit

VOLUME injection
mould

TODAY

a circuit board

PROTOTYPE fab order + hand
assembly + borrowed bench

VOLUME fab / EMS

PROTOPUS

a circuit board

PROTOTYPE Proto + Baker:
finished board on the desk

VOLUME fab / EMS,
with checked files

Plastic has had a desktop prototyping path for years. Electronics hasn't had one that ends in a finished, soldered board.

BOARDS A YEAR → RIGHT TOOL

10

baker

same-day loops beat fab minimums

100

baker

one machine covers a busy lab

1,000

baker + a fab

any batch of 10+ identical boards → fab

10,000

fab / EMS

production; Proto exports checked files

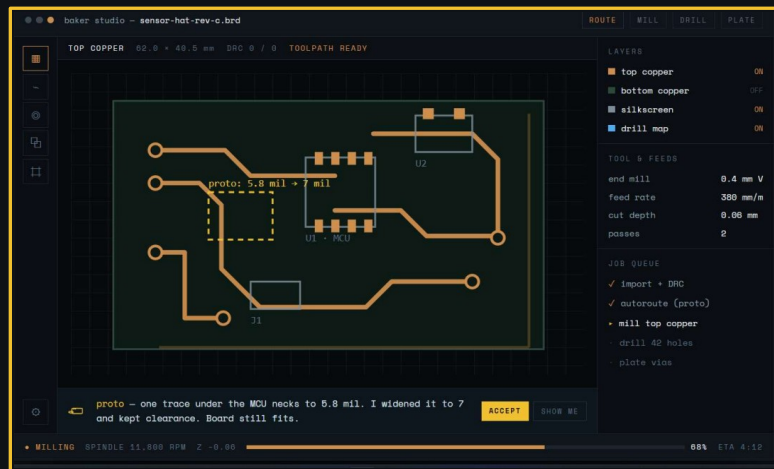
1,00,000

fab / EMS

not what Baker is for

Not a desktop mill: a mill automates one step for experts (Bantam Tools stopped selling its PCB mill in 2025). Baker runs 15 steps, with Proto guiding.

An AI engineer that understands your board



Baker Studio today: Proto found the thin trace, fixed it and planned the mill. You press Accept.

01

READS

schematics and layouts, via the Flux.ai API

02

MAKES

the Gerbers and the Baker job: mill, drill, mask, stencil, jig

03

DIAGNOSES

through Mixer (step 2): waveforms, in plain words

04

HAS A CHARACTER

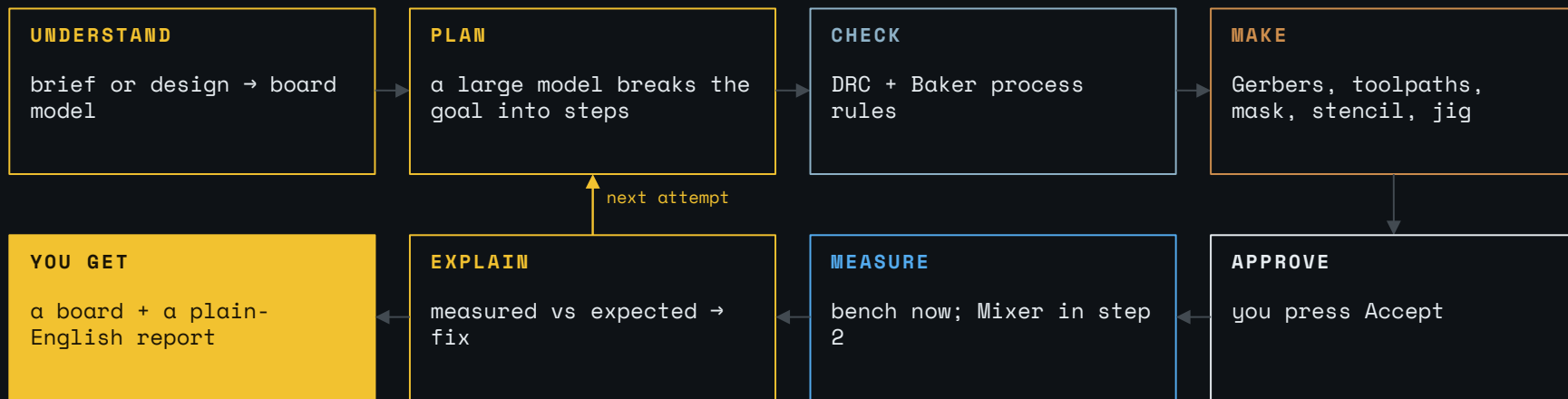
a platypus engineer, built on Gemini

NEXT

Probe Matrix: probes under the board reach every net.

Designed so a **first-timer** can go from idea to board, with Proto at each step.

A loop, not a chatbot



MEMORY every run logs design → settings → measurements → outcome. File-only AI tools never see the real board, so they can't collect this data.

The model proposes · tested engines compute · you approve · the machine executes.

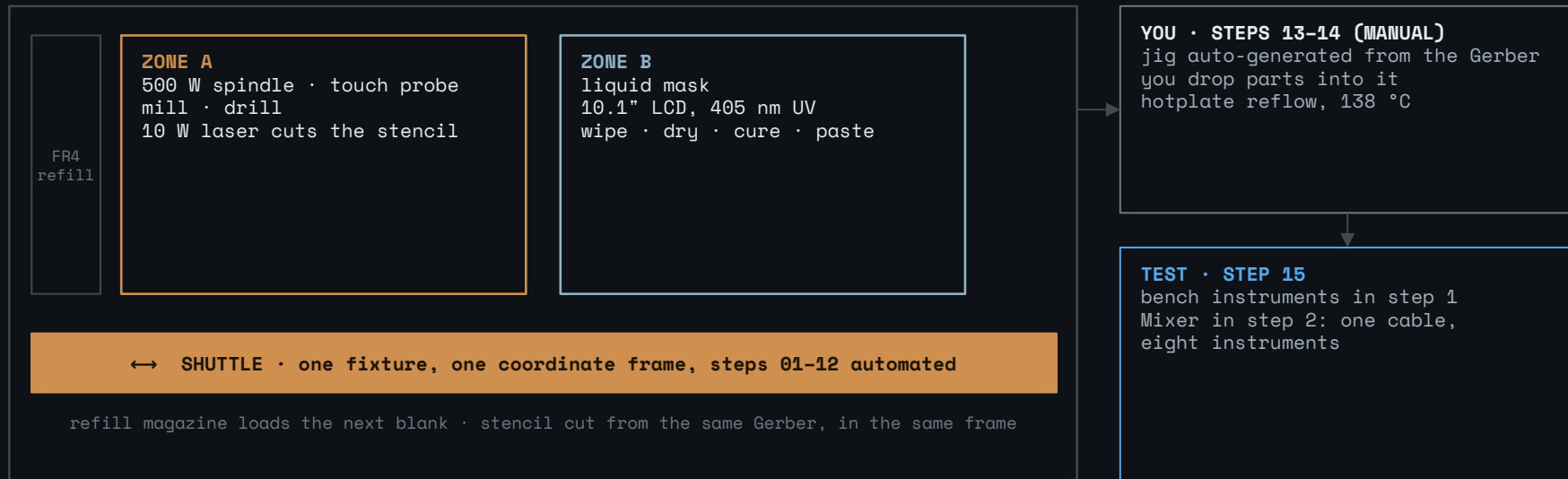
From “import” to “PASS” in one sitting

baker studio — sensor-hat-rev-c.brd

10:02	you	import sensor-hat-rev-c from Flux
10:03	proto	one trace under the MCU necks to 5.8 mil. Widened to 7 mil. [Accept]
10:04	baker	mill · drill 42 holes · mask · stencil · jig prints in parallel
11:40	you	drop parts into the jig
11:52	baker	low-temperature reflow
12:05	mixer	power up ✓ 3V3 rail ✓ I ² C sensor: no reply ✗
12:08	proto	“SDA has no pull-up: R4 is lifted on one pad. Reheat R4.”
12:15	mixer	retest: sensor replies ✓ PASS

Illustrative run of the full system (Mixer is step 2). Times are engineering targets; the proof of concept measures them.

Fifteen steps on one shuttle, in one box



≈2 h

per board (target)

15–20 min

of it hands-on (target)

3–5

boards per shift (target)

₹1.45L

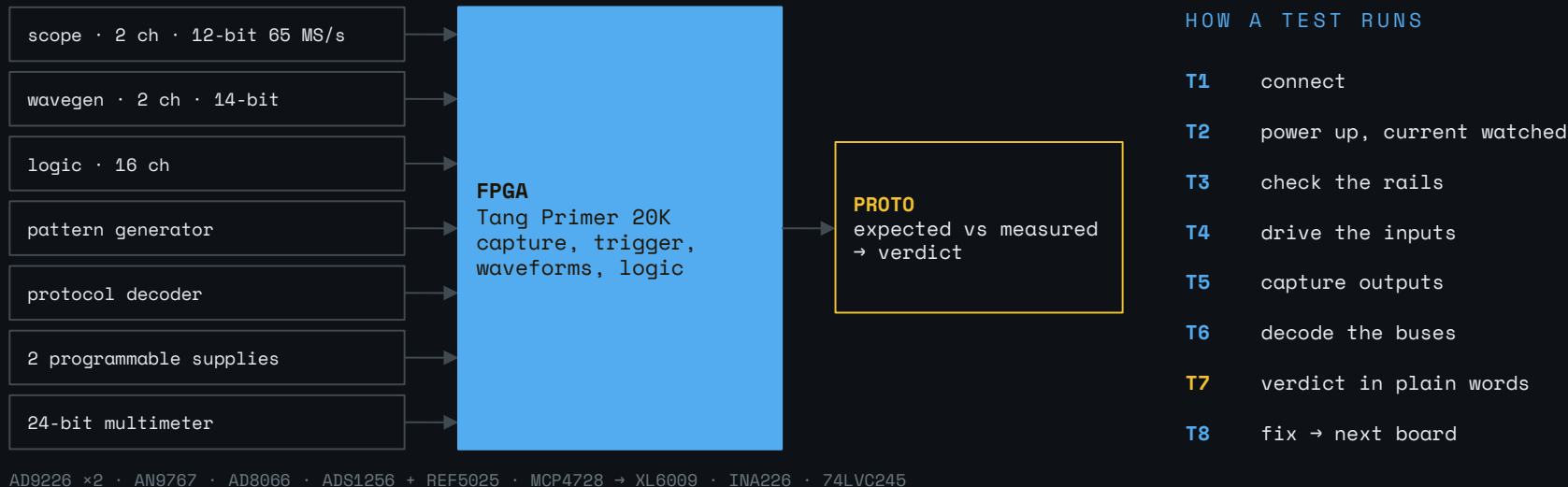
production cost (projected)

Prototype ₹3,73,418 → production ₹1,44,842 (China parts, Delhi build, 100 units) · protopus-kappa.vercel.app/#/baker

Same 15 steps, cheaper parts: ₹3,73,418 → ₹1,44,842

FUNCTION	PROTOTYPE (INDIA, 1 UNIT)	PRODUCTION (CHINA PARTS, 100)	WHY
Axes	Finished ball-screw linear stages	SFU1605 ball screws + HGR15 rails	≈₹5,500 per axis vs ₹16,500
Z axes	100 mm SFU1605 stage + NEMA17 ×2	T8 screw + linear guide	Spindle/laser Z and dispense Z
Control	Raspberry Pi 5 + BTT Octopus Pro + DM556	BTT Manta M8P V2 + CB1 (8 × TMC2209)	One board, 8 drivers, ≈₹10,800
Power	24 V + 48 V + 12 V supplies	One LRS-350-24 rail (spindle keeps its own)	Fewer supplies, simpler wiring
Spindle + laser	500 W ER11 kit · 10 W 445 nm diode (retail)	Volume-sourced; quotes at 100 units	Biggest cost-down lever
Mask + UV	Syringe LPSM · 10.1" 12K LCD · ChiTu L K1 · 405 nm	Same engine	The LCD also prints jigs in production
Placement jig	Separate Saturn 3 Ultra + wash/cure (TR300, 160 °C)	Resin vat + lift under Baker's own UV screen	Removes the printer (to test)
Reflow	300 W silicone hotplate · SSR · MAX31855 · Sn42Bi58 (138 °C)	Same	Stays under the jig resin's 160 °C
Firmware	Klipper + Moonraker + Proto job runner	Locked factory image; activation stays with us	IP by architecture
Unit cost	₹3,73,418 (incl. 35% duty/courier, 15% contingency)	₹1,44,842 (incl. duty, freight, Delhi assembly)	Hardware target ₹1,50,000

Eight instruments, one cable, one brain



₹13,740

production cost (projected)

8 in 1

Analog Discovery 3 class

Probe Matrix

next: full access to every net

Prototype ₹74,847 keeps 12/14-bit converters; production: Tang Nano 9K, AD9288 8-bit 80 MS/s, DAC908, FT232HL, one USB cable, no screen. Built in step 2, after the PoC.

Others solve one step. We're building the whole loop.

PRODUCT	PRICE · ₹	MAKE	ASSEMBLE	TEST	GUIDE
Voltera V-One (ink printer)	~₹3.1 lakh	●	●	○	○
LPKF ProtoMat	8.6–24.5 lakh	●	○	○	○
Makera Carvera (desktop CNC)	~4.8 lakh	●	○	○	○
Bantam Tools PCB mill	discontinued 2025	●	○	○	○
Digilent Analog Discovery 3	~₹34,000	○	○	●	○
JLPCB, PCBWay → India	per order, 7–20 days	●	●	○	○
Flux (AI design)	1.3–4.3k / month	○	○	○	●
protopus · proto + baker	2.99 lakh (+₹34,999 mixer)	●	●	●	●

List prices at ₹88 per US\$, before import duty: Voltera \$3,499.99 · Makera \$5,499 · Digilent \$389 · Flux \$15–49/editor/month · LPKF via Design World · ToolGuyd · JLPCB + zbotic.in. Protopus prices projected.

We earn on every machine. The customer earns it back in months.

PER UNIT · ₹	BAKER	MIXER
Total cost (all-in)	2,00,000	22,500
Selling price	2,99,000	34,999
Margin after all costs	~33%	~36%

₹9,999 / year · one Proto plan per workspace

+ consumables, reordered as labs build · gross margin on build cost: Baker 51.6%

FOR THE CUSTOMER

₹334

consumables per board on Baker

₹6,721

same board from JLCPCB, landed in India

47 boards

to pay back Baker: about 5.9 months for a startup iterating twice a week (11.8 months at once a week). 53 boards once Mixer is added.

Start in the maker studios, then follow the people who watch them

₹6.4L cr global PCB market	₹3.1L cr global test and measurement	₹55K cr India PCB, ~15% a year	₹8.3K cr India test and measurement
33M+ active Arduino users	2.3M+ Hackster hardware community	10,000 Atal Tinkering Labs, +50,000 planned	~2,500 Fab Labs in 125 countries

NOW

Maker studios

Baker units for featured videos. The Bambu Lab playbook.

NEXT

Hobbyists and startups

Everyone who watches and loses weeks per board.

ALONGSIDE

Prototyping labs

Tinkering, Fab and college labs that already buy tools.

Told honestly, one milestone at a time

Baker

prototype running on the bench

Software

Baker Studio + Proto; website live

Mixer

step 2: designed, built after the PoC



SIGNAL

Samuel Beek, Founder & CEO of Schematik, is interested in funding us at proof of concept.

Baker first, one product at a time. Built in the university lab, tested with students and makers. A demo video every month.

₹15,00,000

Baker first: nine months to a proof of concept, and time to raise.

₹3.73L

Baker build, costed
line by line

₹1.60L

Rev B + test-board
consumables

₹4.50L

one software +
firmware engineer, 9
months

₹1.35L

Proto software +
compute

₹1.50L

testing, tooling,
lab safety

₹2.32L

IP, outreach,
reserve

One product at a time: Mixer and Probe Matrix are step 2, funded after the proof of concept. No founder salary. Builds line by line from the cost report.

Step by step: prove Baker, then add the instruments, then the ecosystem

STEP 1 · NOW · ₹15L

Proto + Baker

Proof of concept: 20 boards, measured cycle time, Proto from brief to Baker job.

STEP 2 · SEED · ~₹1.5 CR

Mixer + Probe Matrix

Test and program every board. Baker pilots through China: 20 Bakers, 100 Mixers.

STEP 3 · ECOSYSTEM

Proto runs it all

Any 3D printer joins for enclosures and jigs. India first, then global.

HOW WE RUN THE COMPANY

Software team

improves Proto

Hardware team

machines + accessories

China, turnkey

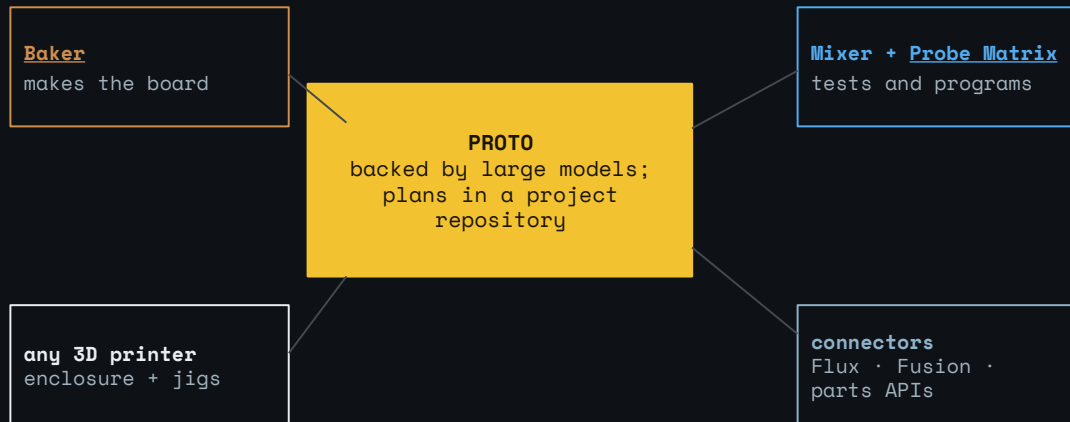
parts, assembly, factory test

India

test, minimal assembly, ship

Seed use (indicative): China pilots + first batches ₹70L · set-up + IP ₹25L · team of 5 ₹25L · Delhi centre ₹20L · buffer ₹10L. Pilot batches are sellable (*₹95L at list).

One plain-English brief in. A tested prototype out.



We never sell a 3D printer: Proto drives the one you have.

The one-stop solution for prototyping: **idea to innovation, without the friction.**

EXAMPLE · A SMARTWATCH WITH 10 BUTTONS

- 01 you describe it in plain English
- 02 Proto checks feasibility and limits
- 03 plans electronics, sensors, PCB; BOM with buy links
- 04 Baker makes the board; the printer prints a case sized from the measured PCB
- 05 Mixer + [Probe Matrix](#) program and test; failing parts are flagged
- 06 you say what felt off; Proto plans the next revision



Thank you.

Come and try the loop.

protopus-kappa.vercel.app

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MANAGEMENT TEAM & CORE SKILLS

Who is building this

Soham

Founder

Third year ECE, Faculty of Technology, University of Delhi

Industrial design

Mechanical architecture

Embedded electronics

Machine software

HIRING NEXT · a software engineer who also writes firmware.

